12.296

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Question

The area enclosed between the straight line $y = x$ and the parabola $y = x^2$ in the xy-plane is (PI 2012)

1. $\frac{1}{6}$
2. $\frac{1}{4}$
3. $\frac{1}{3}$
4. $\frac{1}{2}$

Solution

For $y = x^2$, we write it in matrix form as

$$\mathbf{x}^\top V\mathbf{x} + 2\mathbf{u}^\top\mathbf{x} + f = 0 \tag{1}$$

where

$$V = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} 0 \\ -\frac{1}{2} \end{pmatrix}, \quad f = 0. \tag{2}$$

The line $y = x$ can be written parametrically as

$$\mathbf{x} = \mathbf{h} + \kappa\mathbf{m}, \quad \kappa \in \mathbb{R} \tag{3}$$

where

$$\mathbf{h} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{m} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \tag{4}$$

The intersection points satisfy

$$\kappa_i = \frac{-\mathbf{m}^\top(V\mathbf{h} + \mathbf{u}) \pm \sqrt{[\mathbf{m}^\top(V\mathbf{h} + \mathbf{u})]^2 - (\mathbf{m}^\top V\mathbf{m})\, g(\mathbf{h})}}{\mathbf{m}^\top V\mathbf{m}} \tag{5}$$

where

$$g(\mathbf{h}) = \mathbf{h}^\top V\mathbf{h} + 2\mathbf{u}^\top \mathbf{h} + f. \tag{6}$$

Substituting values:

$$V\mathbf{h} + \mathbf{u} = \begin{pmatrix} 0 \\ -\frac{1}{2} \end{pmatrix}, \tag{7}$$

$$\mathbf{m}^\top (V\mathbf{h} + \mathbf{u}) = -\tfrac{1}{2}, \tag{8}$$

$$\mathbf{m}^\top V\mathbf{m} = 1, \tag{9}$$

$$g(\mathbf{h}) = 0. \tag{10}$$

Hence,

$$\kappa_{1,2} = \frac{-(-\frac{1}{2}) \pm \sqrt{(-\frac{1}{2})^2 - 1(0)}}{1} \tag{11}$$

$$= \frac{\frac{1}{2} \pm \frac{1}{2}}{1}. \tag{12}$$

Solution

Therefore,

$$\kappa_1 = 0, \quad \kappa_2 = 1. \tag{13}$$

The intersection points are

$$\mathbf{x}_1 = \mathbf{h} + \kappa_1 \mathbf{m} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{x}_2 = \mathbf{h} + \kappa_2 \mathbf{m} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \tag{14}$$

Solution

The area enclosed between the line and the parabola is

$$A = \int_0^1 \left(y_{\text{line}} - y_{\text{parabola}}\right) dx \tag{15}$$

$$= \int_0^1 (x - x^2)\, dx \tag{16}$$

$$= \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 \tag{17}$$

$$= \frac{1}{2} - \frac{1}{3} \tag{18}$$

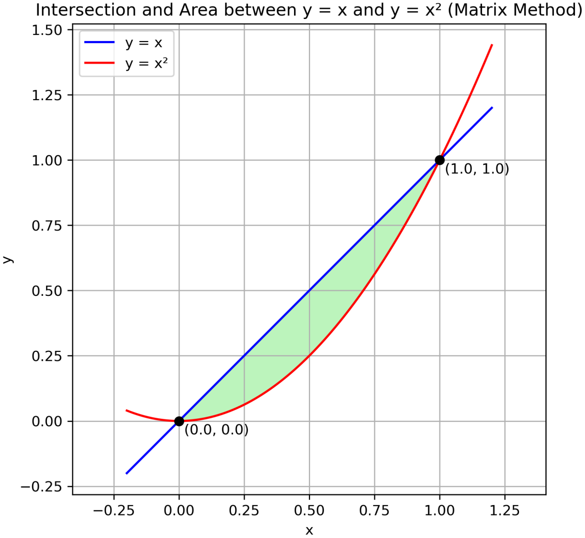
$$= \frac{1}{6}. \tag{19}$$

Solution

$$\boxed{A = \frac{1}{6}} \tag{20}$$

Final Answer: (a) $\frac{1}{6}$

Plot


Intersection and Area between y = x and y = x² (Matrix Method)
y = x
y = x²
(1.0, 1.0)
(0.0, 0.0)
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.25
0.00
0.25
0.50
0.75
1.00
1.25
x
y